CHE 109: Engineering Chemistry – I

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Nanochemistry

Chapter 5

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Nanoscale Materials in Chemistry covers a broad area of science and engineering at the core of future technological development. In particular, the challenges of energy and sustainability are certain to be interrelated with breakthroughs in this area. Among current buzz words (i.e. green, bio-, eco-), "nano" has been used (and abused) to describe an amazingly broad spectrum of systems that has led to frustration for many scientists. The National Nanotechnology Initiative has defined nanotechnology as "working at the atomic, molecular and supramolecular levels, in the length scale of approximately 1–100 nm range, in order to understand and create materials, devices and systems with fundamentally new properties and functions because of their small structure" (www.nano.gov). Naturally, this broadly defined area of science and engineering has a significant "chemistry" component. This book

- Nano science and technology is a broad and interdisciplinary area growing explosively worldwide in the past few years. Nanomaterials are cornerstones of nanoscience and nanotechnology.

- The science and technology which deals with the particles in size between 1 to 100nm is known as nano science and nano technology.

Classification of nanomaterials on the basis of dimensions

S. No.	Reduction in size in different coordinates	Size	Examples
1.	3-dimensions	< 100 nm	Nanoparticles, quantum dots
2.	2-dimensions	< 100 nm	Nanotubes, nanowires, nanofibers
3.	1-dimension	< 100 nm	Thin films, coatings

Classification based on pore dimensions

- A useful way to classify nanoporous materials is by the diameter size of their pores, since most of the properties, which are interesting for the applications of adsorption and diffusion are dependent on this parameter.

Microporous materials ($d<2$ nm): These materials have very narrow pores. They can host only small molecules, such as gases or linear molecules, and generally show slow diffusion kinetics and high interaction properties. They are generally used in gas purification systems, membrane filters or gas-storage materials.

Example: Na-Y and naturally occurring clay materials.

Mesoporous materials ($2<d<50$ nm): These materials have pores with diameter size enough to host some big molecules, for example aromatic systems or large polymeric monomers. Diffusion kinetics of the adsorbed molecules is often due to capillarity, with an initial interaction with the pore wall followed by pore filling. These systems can be used as nano-reactors for the polymerization or adsorbing systems for liquids or vapours.

Example: MCM-41, MCM48, SBA15 and carbon mesoporous materials etc.

Macroporous systems ($d>50$ nm): Pores of these materials could host very large molecules, such as poly-aromatic systems or small biological molecules, and interactions with pore walls are often secondary respect to the interactions with other molecules, overall in the case of very small guest molecules. These materials are principally used as matrices to store functional molecules, as scaffolds to graft functional groups, such as catalytic centres, and as sensing materials, thanks to the quick diffusion of chemical species in the pore system.

Example: Carbon micro tubes, Porous gels and porous glasses

Classification of nanomaterials based on origin

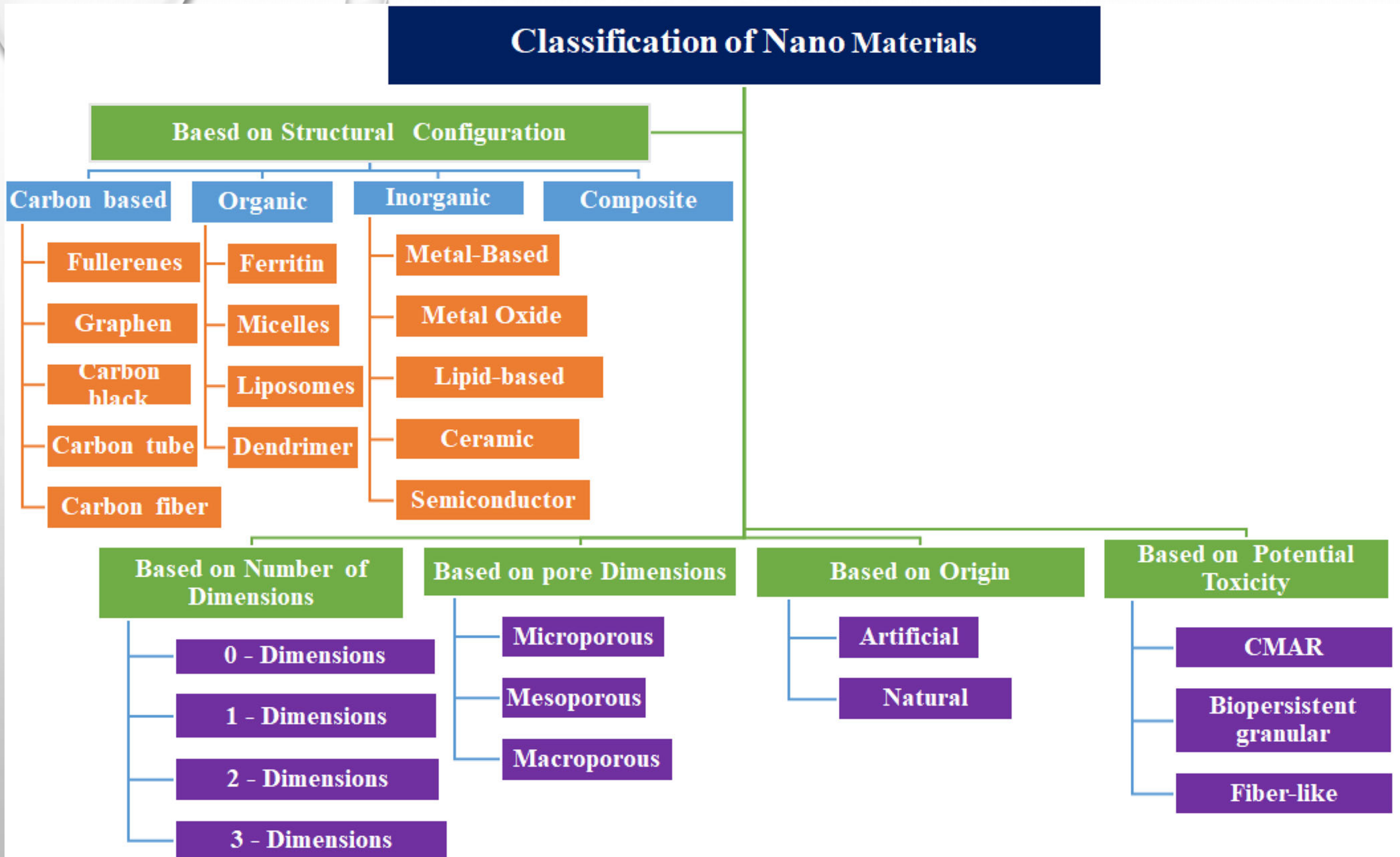
Natural and artificial nanoparticles are the two groups into which nanomaterials are divided based on origin.

Natural nanomaterials

Natural nanomaterials can be found in a variety of forms in nature, including viruses, protein molecules, minerals like clay, natural colloids like milk and blood (liquid colloids), fog (aerosol type), gelatin (gel type), mineralized natural materials like shells, corals, and bones, insect wings and opals, spider silk, lotus leaves, gecko feet, volcanic ash, and ocean spray.

Artificial nanomaterials

Carbon nanotubes and semiconductor nanoparticles like quantum dots (QDs) are examples of artificial nanomaterials that are made consciously using precise mechanical and manufacturing procedures. Nanomaterials are categorized as metal-based materials, dendrimers, or composites depending on their structural makeup.


Classification of Nano Materials
Baesd on Structural Configuration
Carbon based
Organic
Inorganic
Composite
Fullerenes
Graphen
Carbon black
Carbon tube
Carbon fiber
Ferritin
Micelles
Liposomes
Dendrimer
Metal-Based
Metal Oxide
Lipid-based
Ceramic
Semiconductor
Based on Number of Dimensions
0 - Dimensions
1 - Dimensions
2 - Dimensions
3 - Dimensions
Based on pore Dimensions
Microporous
Mesoporous
Macroporous
Based on Origin
Artificial
Natural
Based on Potential Toxicity
CMAR
Biopersistent granular
Fiber-like

Synthetic approaches of Nanomaterials

(i) Bottom-up approach: The building of nanostructures starting with small components such as atoms or molecules is called bottom-up approach.

Ex: Chemical vapour deposition, Sol-Gel Process, Chemical Reduction methods, etc.

(ii) Top-down approach: The process of making nanostructures starting with larger structures and breaking away to nano size is called top-down approach.

Ex: Lithography, Ball milling, Epitaxy, etc.

Characterization techniques of nanomaterials

S. No.	Techniques	Information acquired
1.	Scanning Electron Microscopy (SEM) with Energy-dispersive X-ray spectroscopy	Surface topography (up to 10nm) and composition
2.	Transmission Electron Microscopy (TEM)	Surface morphology (up to 0.2nm)
3.	Atomic Force Microscopy	Identification of individual surface atoms
4.	Particle Size Analyzer	Particle Size distribution
5.	FT-Raman Spectra	Distinguish single walled carbon nanotubes and multi walled carbon nanotubes
6.	Photoluminescence Spectra	CNT chirality or Asymmetry determination
7.	X-ray photoelectron spectroscopy	Electronic state of the element

PROPERTIES OF NANOMATERIALS

The various properties, which get tremendously altered due to the size reduction in at least one dimension are:

a) Chemical properties: Reactivity; Catalysis.
b) Thermal property: Melting point temperature.
c) Electronic properties: Electrical conduction.
d) Optical properties: Absorption and scattering of light.
e) Magnetic properties: Magnetization.

Chemical Properties

- Based on the surface area to volume effect, nanoscale materials have:

  a) Increased total surface area.

  b) Increased number of atoms accessible on the surface.

  c) Increased catalytic activity of those large number surface atoms.

  d) Different/tunable surface catalytic properties by the change in shape, size and composition.

- Hence, nanoscale catalysts can increase the rate, selectivity and efficiency of various chemical reactions.

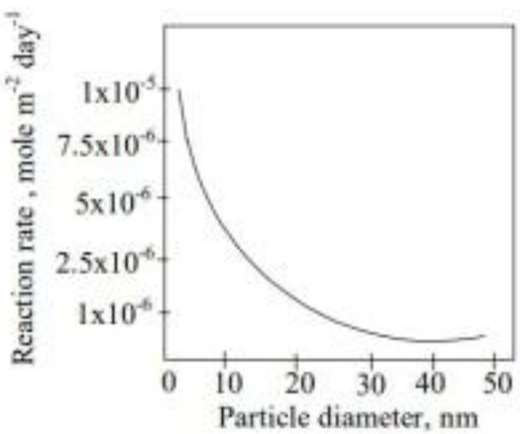


Fig 1.1: Effect of Particle Size on the Reaction Rate

Thermal Properties

- The melting point of a material directly correlates with the bond strength. In bulk materials, the surface to volume ratio is small and hence the surface effects can be neglected. However, in nanomaterials the melting temperature is size dependent and it decreases with the decrease particle size diameters.

- The reason is that in nanoscale materials, surface atoms are not bonded in direction normal to the surface plane and hence the surface atoms will have more freedom to move.

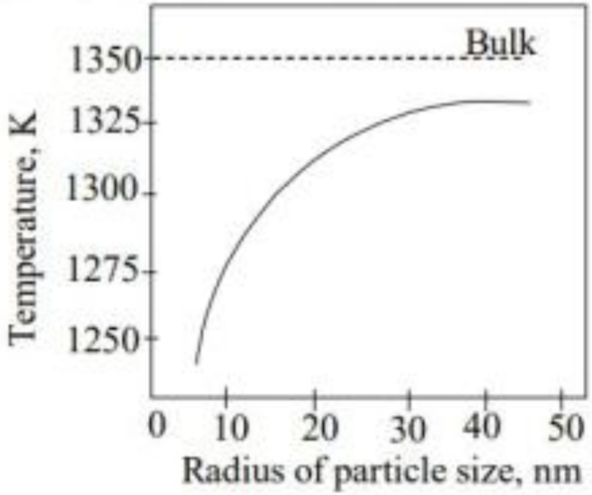


Fig 1.2: Effect of Particle Size on the Melting Point

Electronics Properties

- In bulk materials, conduction of electrons is delocalized, that is, electrons can move freely in all directions.
- When the scale is reduced to nanoscale, the quantum effect dominates. For zero dimensional nanomaterials, all the dimensions are at the nanoscale and hence the electrons are confined in 3-D space. Therefore no electron delocalization (freedom to move) occurs.
- For one dimensional nanomaterials, electrons confinement occurs in 2-D space and hence electron delocalization takes place along the axis of nanotubes/nanorods/nanowires.
- Due to electron confinement, the energy bands are replaced by discrete energy states which make the conducting materials to behave like either semiconductors or insulators.

Synthesis of Nanomaterials

1.4 NANOPARTICLES

Particles or powders with particle size less than 100nm are called nanoparticles.

1.4.1 Synthesis of Nanoparticles by Chemical Reduction Method

Group VIB metal halides like $CrCl_3$, $MoCl_3$, WCl_4 can be reduced into their corresponding metals by using $NaBEt_3H$ (sodium triethoxy boron hydride) with toluene as the solvent at room temperature.

$$MCl_3 + \ xNaBEt_3H \xrightarrow{Toluene} M + xNaCl + xBEt_3 + {}^{x}/_{2}\,H_2$$

$$CrCl_3 + \ 3NaBEt_3H \xrightarrow{Toluene} Cr + 3NaCl + 3BEt_3 + {}^{3}/_{2}\,H_2$$

Properties of Nanomaterials

(i) As the particle size decreases, surface area increases. This enhances the catalytic activity of the nanoparticles.

(ii) Reduction of particle size from micron to nanometer scale influences their optical properties.

For Example: CdS in micron size appear as red, 6nm size appear orange in colour, 4nm size is yellow coloured and 2nm size appear as white.

(iii) Reduction of particle size from micron to nanometer scale influences the thermal properties like melting point and thermal conductivity.

Application of Nanomaterials

a) Silver nanoparticles have good antibacterial properties, and are used in surgical instruments, refrigerators, air-conditioners, water purifiers etc.

b) Gold nanoparticles are used in catalytic synthesis of silicon nano wires, sensors carrying the drugs and in the detection of tumors.

c) ZnO nanoparticles are used in electronics, ultraviolet (UV) light emitters, piezoelectric devices and chemical sensors.

d) TiO_2 nanoparticles are used as photocatalyst and sunscreen cosmetics (UV blocking pigment).

e) Antimony-Tin-Oxide (ATO), Indium-Tin-Oxide (ITO) nanoparticles are used in car windows, liquid crystal displays and in solar cell preparations.

Synthesis of Nanoporous materials by Sol-gel process

Sol-Gel Process:

"Formation of an oxide network through polycondensation reactions of a molecular precursor in a liquid".

Precursors

The precursor used in sol-gel process for the synthesis of nanoporous materials are metal alkoxides $M(OR)_4$. They readily react with water to form gels.

Examples

- Tetra methoxy silane [$Si(O_3CH_4)$]
- Tetra ethoxy silane [$Si(O_2C_5H_4)$]
- Tetra butoxy titanate [$Ti(O_4C_9H_4)$]

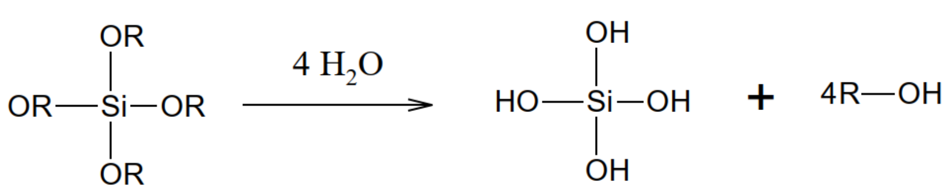
Process (Synthesis of silica aerogel)

This process consists of four main steps.

1. Hydrolysis of precursors
2. Condensation followed by polycondensation
3. Gelation
4. Supercritical drying

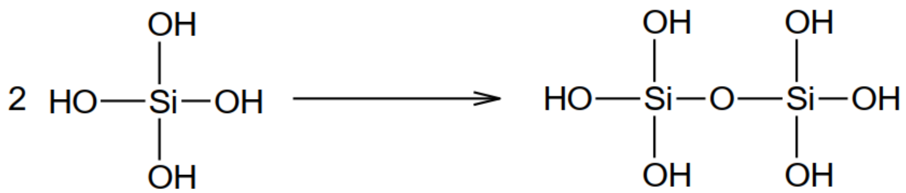
1. Hydrolysis

It occurs by the addition of water to any one of the precursor material to form silanol (Si-OH) particles.



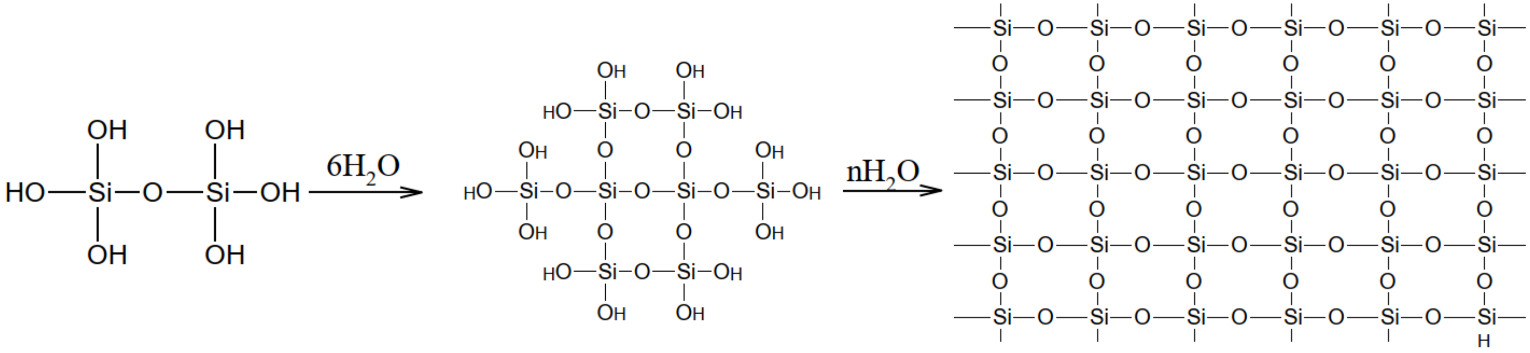
2. Condensation

The self condensation of silanol groups produces siloxane linkages filled with by products of water and alcohol.



3. Polycondensation

The condensation process continues to form poly condensed silica gel with Si-O-Si linkages.



4. Drying

The gels are subjected to super critical drying in an autoclave. The critical pressure and critical temperature used are 78 bar and 294^{o}C respectively in order to remove liquid from silica gel to form the network structure of silica aerogel.

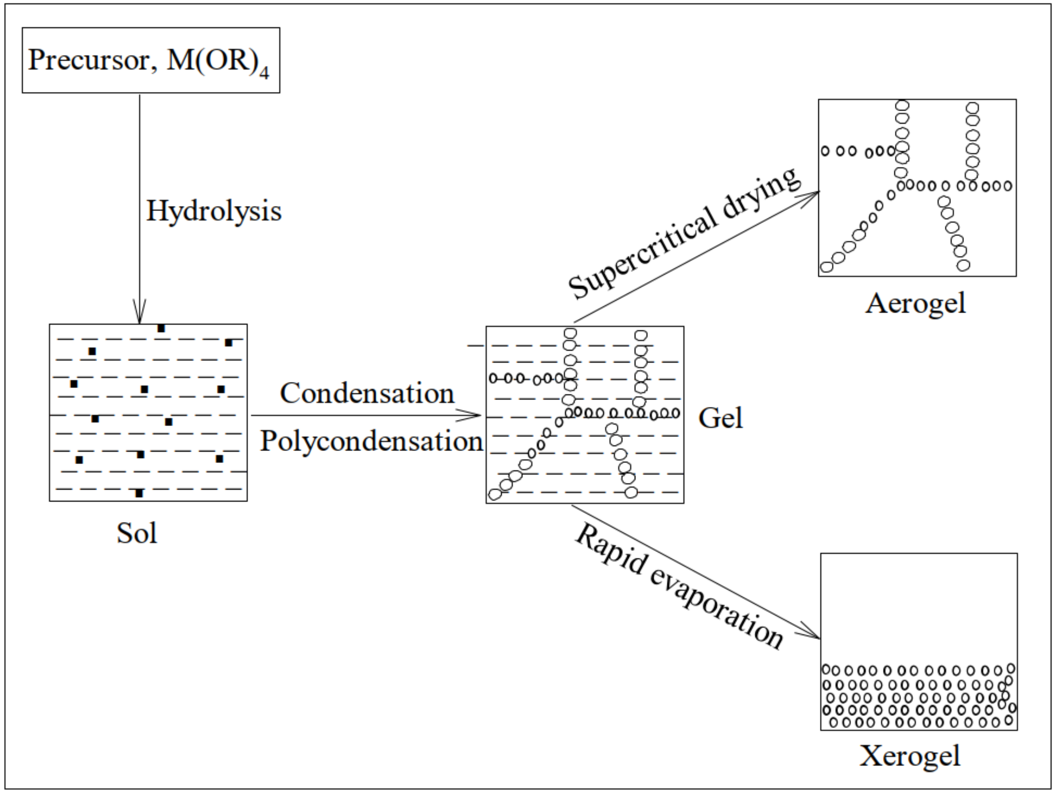


Fig. 1.6: Sol-gel process

Applications of Sol-gel process

- It can be used in ceramics manufacturing processes as an investment casting material or as a means of producing very thin films of metal oxides for various purposes.
- Sol-gel derived materials have diverse applications in optics, electronics, energy, space, (bio) sensors, medicine (e.g. controlled drug release) and separation (e.g. chromatography) technology.
- One of the more important applications of sol-gel processing is to carry out zeolite synthesis.

- Other elements (metals, metal oxides) can be easily incorporated into the final product and the silicalite sol formed by this method is very stable.
- Other products fabricated with this process include various ceramic membranes for microfiltration, ultrafiltration, nanofiltration and reverse osmosis.